

Shadowing Assessment #1

Date of Shadowing: November 19, 2024

Professional Observed: Dr. Anup Tilak and lab staff

Location: Purepath Lab & Therapeutics

Introduction:

This shadowing experience stemmed from my previous interview with Dr. Anup Tilak (see Interview Assessment #2), where I was invited to observe the diagnostic lab operations firsthand. During this first session, I was introduced to the foundational molecular biology procedures and the different solutions and equipment used for DNA/RNA extraction and sample processing.

Observations:

- Learned about the preparation of elution samples using DEPC-treated ddH₂O.
- Observed the sample plate (mye plate) preparation:
 - 500 µL of isopropanol
 - 10 µL of Proteinase K
 - 2 µL of L solution
 - 250 µL of Buffer Mye
 - 250 µL of the biological sample
- Understood the mixture of magpure beads (10 µL per sample + 190 µL storage solution ddH₂O) for sample processing.
- Discussed COVID testing procedures using V-NAT buffer and the role of positive and negative controls in ensuring test reliability.
- Noted the use of synthetic controls for RPP (Respiratory Pathogen Panel) testing.
- Understood the differences between DNA and RNA extraction methods and how they are critical in molecular diagnostics.
- Learned about preventive maintenance vendors like Nugget Scientific.

Reflection:

This session was an eye-opener into how much detail and precision goes into each molecular biology procedure. The complex process of sample preparation, reagent selection, and equipment calibration made me appreciate the expertise required in diagnostic labs. Dr. Tilak's lab uses a variety of reagents and steps that emphasize safety and accuracy.

Skills/Knowledge Gained:

- Molecular biology techniques in diagnostic testing
- Components of sample preparation
- Use of positive and negative controls

- Understanding PCR test setup and controls